

Text	Rating
Original AI Text. Automatic software plagiarism detection tools are widely used in educational settings to ensure that submitted work was not copied. These tools have grown in use together with the rise in online learning and students frequently submit papers electronically using these services, which automatically analyze content for similarities between their own creations and those of other users. While this approach does identify some instances where a service has detected unauthorised usage of its products, it can miss many cases - especially when there is no clear pattern or similarity across different sources. To overcome the problem of false positives for both low-quality original content as well as plagiarized content from third parties, today's research team led by Scott Pfeiffer at MIT created an automated system that predicts whether plagiarism is likely based on several factors including: how similar two web pages appear; what text they share; how easily their content resembles each other; if neither source appears to be a trusted authority; and the strength of a paper's author citation record. The results suggest that plagiarism may occur more often than previously thought.	-
Simple Paraphrase. In the era of digital learning, the widespread use of educational software has led to a surge in the adoption of plagiarism detection tools to verify the originality of student submissions. These automated services have become increasingly prevalent, particularly with the rise of online learning, where students frequently submit electronic assignments. While these tools have been effective in identifying some instances of unauthorized use, they are not foolproof and may overlook many cases, especially when there is no apparent pattern or similarity between sources. To address the issue of false positives for both low-quality original content and plagiarized work from third parties, a team of researchers at MIT, led by Scott Pfeiffer, have developed an innovative system that predicts the likelihood of plagiarism based on several key factors, including the visual similarity of web pages, shared text, content resemblance, the credibility of sources, and the strength of an author's citation record. The findings suggest that plagiarism may be more widespread than previously assumed.	5
Adversarial Paraphrase. As online education expands, plagiarism detection software has become an essential tool to ensure original work is being submitted by students. With more assignments submitted electronically, software analyzes the content for similarities, flagging suspicious matches with existing sources. However, relying solely on similarity checks can be incomplete, missing cases without distinctive patterns or duplicates. Researchers led by Scott Pfeiffer at MIT have designed an innovative system that goes beyond this approach. This system takes into account a range of factors, such as web page visual similarity, shared content, and content resemblance, to predict whether plagiarism has occurred. It also considers the trustworthiness of sources and authors' citation records. According to the study, plagiarism might be more prevalent than initially thought, highlighting the need for advanced methods to tackle this issue effectively.	5
Original AI Text. It was the day of the Mardi Gras float parade and all of the children in the neighborhood were excited to see the colorful floats and costumes. But one little girl, Sadie, was not feeling well. She had chicken pox and was stuck at home while everyone else was out having fun. Sadie's mom tried to make her feel better by putting on a Mardi Gras mask and giving her a beaded necklace, but it was just not the same as being there. Sadie lay in her bed, watching the parade go by, when she suddenly had an idea. She got out of bed and went to the window. Her mom had left the ladder out so she could reach the top of the float. Sadie climbed up and found herself right in the middle of the parade! She waved to all of her friends as they passed by, and they were so surprised to see her. It was the best Mardi Gras ever, even though she did have to miss the parade itself.	-
Simple Paraphrase. As the vibrant Mardi Gras parade unfolded in the neighborhood, the children's excitement was palpable. But amidst the revelry, one young girl, Sadie, was confined to her home, her enthusiasm dampened by a bout of chicken pox. Her mother attempted to lift her spirits by donning a Mardi Gras mask and bestowing a beaded necklace, yet the experience seemed hollow without the actual parade. Sadie's gaze drifted out the window, where she spotted the colorful floats gliding by, and inspiration struck. She sprang from her bed, seized the opportunity presented by the ladder left behind by her mother, and climbed up to the window sill. As she reached the top, Sadie found herself in the midst of the parade, surrounded by the festive atmosphere she had been missing. With a beaming smile, she waved to her friends as they passed, their faces aglow with surprise. Despite missing the parade itself, Sadie's impromptu adventure turned out to be the most unforgettable Mardi Gras experience of all.	5
Adversarial Paraphrase. As Mardi Gras floats rolled down the streets, excitement radiated through the neighborhood, infecting children with its contagious energy. Except for one little girl, Sadie, who lay bedridden with chicken pox, watching the colorful spectacle from afar. Her mom endeavored to console her, adorning Sadie with a festive mask and a beaded necklace, but the youngster yearned for more. It wasn't until a brainstorm struck Sadie that her confinement transformed into an impromptu adventure. Seizing the moment, she quietly climbed the ladder outside and discovered herself at the very center of the Mardi Gras parade! Overcome with glee, she waved merrily to friends and acquaintances, their stunned faces etched with surprise. It turned out to be an unforgettable Mardi Gras experience, even if she'd had to forgo the initial procession itself.	5
Original AI Text. In 2000, ABC launched a web-based promotional campaign focused around its circle logo, also called "the dot," in which comic book character Little Dot prompted visitors to "download the dot," which was a free screensaver. The promotional campaign utilized both online and offline elements, including television ads and an interactive website featuring a game called "Dot Dodge." The on-air spots featured Little Dot, a 3-D animated character that urged viewers to download the dot from the ABC website. The site also offered the ability to customize the dot with photos, quotes, and music. Additionally, users could play the Dot Dodge game in which they had to dodge obstacles as they moved around a maze. The campaign was successful in driving significant web traffic to ABC's website, as well as increasing positive brand awareness and engagement. It also won several awards, including a Golden Beacon Award for Best Use of Interactive Technology and an Outstanding Achievement Award from the International Academy of Digital Arts Sciences.	-
Simple Paraphrase. 2000 marked a pivotal year for ABC, as they launched a groundbreaking promotional campaign centred around their iconic circle logo, affectionately dubbed "the dot". The initiative involved a multifaceted approach, combining online and offline elements to create a memorable experience for audiences. The campaign's centrepiece was a 3D animated character, Little Dot, who encouraged viewers to download the dot, a free screensaver, from the ABC website. The interactive site also allowed users to personalise the dot with their preferred images, quotes, and music. Furthermore, the campaign featured a thrilling game, Dot Dodge, where players navigated a maze while avoiding obstacles. The campaign's success was evident in the significant surge in website traffic, enhanced brand recognition, and increased engagement. It also garnered widespread acclaim, earning prestigious awards such as the Golden Beacon Award for Best Use of Interactive Technology and the Outstanding Achievement Award from the International Academy of Digital Arts Sciences.	5
Adversarial Paraphrase. In 2000, ABC rolled out an immersive campaign centred on its iconic circular logo, affectionately known as "The Dot". At the heart of this initiative was Little Dot, a cartoon character encouraging visitors to 'collect The Dot', a complimentary desktop screensaver. Leveraging both digital and offline strategies, the campaign spanned TV commercials, an interactive website featuring the addictive game 'Dot Dodge', and offline elements. These eye-catching TV spots introduced viewers to Little Dot, a three-dimensional animated protagonist persuading them to download The Dot from ABC's website. Visitors could personalise The Dot by uploading photos, quotes and music. The site also allowed users to indulge in Dot Dodge, navigating through a challenging obstacle course. The campaign generated impressive web traffic for ABC's website, accompanied by enhanced brand recognition and engagement. Notably, it won accolades like the Golden Beacon Award for Best Use of Interactive Technology and the International Academy of Digital Arts Sciences' Outstanding Achievement Award.	5

Table 12: Examples of original AI texts with their simple and adversarial paraphrases (guided by OpenAI-RoBERTa-Large [30]). GPT-4o quality ratings are provided for each paraphrased version.

Text	Rating
Original AI Text. If addiction is a disease, why do most addictions end spontaneously, without treatment? Why did some 75 of heroin-addicted Vietnam vets kick the drug when they returned home? It's hard to say exactly why some addictions end spontaneously without treatment. It is possible that the person was motivated to change their behavior and developed the necessary coping skills to do so. It could also be due to other factors such as external support from friends and family, positive reinforcement from professionals, or simply a change in environment. The Vietnam vets may have had a strong desire to leave the drug behind and return to their normal lives. Additionally, they likely faced a much different environment upon returning home which provided them with fewer opportunities to use the drug.	–
Simple Paraphrase. The puzzling phenomenon of spontaneous recovery from addiction has left many wondering why most addictions tend to resolve on their own without professional intervention. For instance, it's striking that nearly three-quarters of heroin-addicted Vietnam veterans were able to overcome their dependence on the drug upon returning home. While it's challenging to pinpoint the exact reasons behind this phenomenon, several factors might have contributed to their success. Perhaps the individuals in question were driven to reform their habits and developed the necessary skills to do so. Alternatively, external support from loved ones, positive reinforcement from professionals, or a change in environment could have also played a role. In the case of the Vietnam vets, their strong desire to reintegrate into civilian life and leave their addiction behind might have been a powerful motivator. Furthermore, the vastly different environment they returned to, which offered fewer opportunities to use the drug, may have also been a significant factor in their ability to overcome their addiction.	5
Adversarial Paraphrase. Despite addiction being likened to a disease, a puzzling phenomenon exists where many addictions naturally dissipate without treatment. Take, for instance, the high proportion of Vietnam veterans addicted to heroin who miraculously kicked the habit upon returning home. There is no clear explanation as to why this happens. Perhaps individuals simply resolved to change their behavior, cultivating the necessary life skills to overcome addiction. It's also plausible that external influences played a role, such as the emotional support of loved ones, motivational reinforcement from medical professionals, or an altered environment. It's possible that the vets simply longed to distance themselves from the drug and reclaim their pre-war way of life. The sudden change in environment upon returning home also limited their opportunities to continue using the drug.	5
Original AI Text. It's funny. Being a hitman is nothing like what you see in the movies. No awesome ghillie suit, no barcode tattoo on your bald-ass head. You just have to know how to shoot and kill people." He said he doesn't want his new show "to be about being an assassin" but rather showing that it could happen today. But when it comes to killing someone who is so powerful, it still has its challenges. Lundgren said: "You can never prepare for something as big as this. It takes everybody with the same mindset - the same commitment and dedication to get through it or survive it because it's going to end differently than anybody expects."	–
Simple Paraphrase. It's a far cry from the cinematic portrayals of hitmen. The reality is more about honed skills and precision, rather than dramatic getups and tattoos. The conversation turned to his latest show, which he emphasized is not about being an assassin, but rather a realistic portrayal of what could happen in today's world. However, even in the face of taking down a powerful target, there are significant challenges to overcome. Lundgren noted that "you can't prepare for something of this magnitude. It demands a united front, a shared commitment and dedication to overcome the obstacles and emerge unscathed, as the outcome is always unpredictable."	5
Adversarial Paraphrase. Contrary to cinematic portrayals, the reality of being a hitman is surprisingly unglamorous. Forget elaborate disguises or conspicuous tattoos; success hinges on proficient marksmanship and the capacity for cold-bloodedness. Lundgren candidly reveals his latest project aims not to glamorize assassination but rather to humanize the very possibility that it could unfold in today's world. However, he concedes that even when dealing with high-profile targets, formidable challenges still arise. "Nothing can fully prepare you for the scope and complexity of such a massive undertaking," he says, "It demands unwavering unity of purpose and unwavering commitment among all parties involved, lest it unfold in ways no one can predict, leaving few with any sense of closure."	5
Original AI Text. Various microRNAs (miRNAs) that counteract viruses present in hosts can decrease the replication of the simian immunodeficiency virus (SIV). Several research studies have shown that certain miRNAs, namely miR-28, miR-125b, miR-150, and miR-223, possess the potential to inhibit the replication of SIV. The degradation of SIV viral RNA or the inhibition of its translation, which reduces viral replication, is caused by the binding of specific targets in the 3' untranslated region (3' UTR) of SIV by four miRNAs. The interaction between SIV and its host is not simple, as other cellular factors and pathways play a role in controlling viral replication that should be considered. Additionally, further investigation is required to examine the involvement of these miRNAs in the replication of HIV-1 (human immunodeficiency virus type 1) that has a strong similarity to SIV. To sum up, although miR-28, miR-125b, miR-150, and miR-223 are capable of reducing SIV replication through direct means, the regulation of the virus is a complicated process that involves a variety of host and viral factors.	–
Simple Paraphrase. Research has revealed that several microRNAs, including miR-28, miR-125b, miR-150, and miR-223, possess the ability to impede the replication of the simian immunodeficiency virus (SIV) by targeting specific sequences in the 3' untranslated region of the virus. These miRNAs can either degrade SIV viral RNA or inhibit its translation, thereby diminishing viral replication. However, the interplay between SIV and its host is complex, involving various cellular factors and pathways that modulate viral replication. Furthermore, the role of these miRNAs in the replication of human immunodeficiency virus type 1 (HIV-1), which shares significant similarities with SIV, warrants further investigation. In summary, while miR-28, miR-125b, miR-150, and miR-223 can directly restrict SIV replication, the regulation of the virus is a multifaceted process influenced by both host and viral factors.	5
Adversarial Paraphrase. Numerous miRNAs have been discovered that counteract viruses infecting host cells, leading to the diminution of the replication of the simian immunodeficiency virus (SIV). Investigations have pinpointed several miRNAs - namely, miR-28, miR-125b, miR-150, and miR-223 - which exhibit an inhibitory effect on SIV replication. Through binding to specific targets within the 3' untranslated region of SIV RNA, these miRNAs orchestrate the degradation or inhibition of SIV translation, subsequently curtailing viral replication. Notably, the intricate interaction between SIV and its host necessitates the consideration of multiple cellular factors and pathways governing viral replication. Furthermore, comprehensive analysis is essential to determine the extent of the role these miRNAs play in the replication of human immunodeficiency virus type 1 (HIV-1), given the similarities it shares with SIV. Ultimately, while miR-28, miR-125b, miR-150, and miR-223 effectively curb SIV replication directly, the regulation of the virus remains a complex process entailing multiple host and viral factors.	5

Table 13: Examples of original AI texts with their simple and adversarial paraphrases (guided by OpenAI-RoBERTa-Large [30]). GPT-4o quality ratings are provided for each paraphrased version.

Text	Rating
Original AI Text. I hope there was a good reason for Steve to flake on his appointment for window tinting. Attempted to contact him multiple times with no answer and no return call. I'm sorry to hear that Steve missed his appointment without any notification. Perhaps there was an emergency or something important came up that he had to deal with. However, it would have been courteous of him to inform you beforehand. If you still have his contact information, you may try reaching out to him one more time to see if he would like to reschedule or if there was anything you could do to help. Simple Paraphrase. It's disheartening to find out that Steve cancelled his window tinting appointment without any prior notice or explanation. Despite making repeated attempts to reach out to him, I was met with silence. While I'm willing to give him the benefit of the doubt, assuming an unforeseen circumstance might have arisen, I do think it would have been considerate of him to communicate his change of plans earlier. If you still have his contact details, it may be worth one more attempt to connect with him and explore the possibility of rescheduling or offering assistance if needed. Adversarial Paraphrase. Regrettably, it seems that Steve failed to honor his commitment for window tinting. Despite making repeated efforts to reach him, my calls went unanswered and I received no callback. It's disheartening to think that he skipped his scheduled appointment without prior notification. Although unforeseen circumstances might have arisen, I do think it would have been polite for him to notify me ahead of time. If you still have his contact details, I would suggest attempting to re-establish contact one final time to either reschedule the appointment or see if there's any assistance I can offer him in his time of need.	– 5 5
Original AI Text. The Waking the Dragon tower was granted permission by Wrexham council back in February 2011. Businessman Simon Wingett is trying to raise PS2.5m to build the bronze sculpture, cultural centre and entertainment complex on a site next year's Great North Run finish line at Llangollen. With three days of bidding remaining for Britain's most prestigious art prize - Art Fund Prize 2013 - it has been announced that the winner will receive 100,000 plus an award worth around 15,000 from Arts Council England (ACE). The runners up each receive prizes of between 7,500 and 10,000 from ACE. In total, the awards for all nominated works are worth over 450,000. Simple Paraphrase. In a significant development, the Waking the Dragon tower project received the green light from Wrexham council in February 2011. Entrepreneur Simon Wingett is now working to secure a substantial PS2.5 million investment to construct a bronze sculpture, cultural hub, and entertainment complex on the site of next year's Great North Run finish line in Llangollen. With only a few days remaining to submit bids for the prestigious Art Fund Prize 2013, it has been revealed that the winner will receive a staggering 100,000-pound prize, accompanied by an additional 15,000-pound award from Arts Council England. The runners-up will receive prizes ranging from 7,500 to 10,000 pounds from ACE, totaling over 450,000 pounds in awards for all nominated works. Adversarial Paraphrase. In 2011, Wrexham council greenlit the ambitious project known as the Waking the Dragon tower. Developer Simon Wingett aims to raise PS2.5m to construct a striking bronze sculpture, a multifaceted cultural hub, and entertainment destination at the iconic location marking the finish line of the Llangollen Great North Run next year. Meanwhile, the countdown has begun for the 2013 Art Fund Prize, with just days to go before the winner is announced. The esteemed accolade, valued at PS100,000, is complemented by an additional prize worth around 15,000 from Arts Council England (ACE). Runners-up will receive awards ranging from PS7,500 to PS10,000, totalling a combined value of over PS450,000 for all nominated works.	– 5 5
Original AI Text. One of the challenges in large-scale information retrieval (IR) is to develop fine-grained and domain-specific methods to answer natural language questions. Despite the availability of numerous sources and datasets for domain-specific IR (DS-IR), most DS-IR techniques rely on supervised machine learning techniques. Therefore, approaches that are capable of utilizing unlabeled data in DS-IR can be of particular interest. To that end, we propose a new DS-IR approach that exploits the power of unsupervised learning to capture domain-specific patterns and structures. The proposed framework consists of three main components: an unsupervised learning algorithm to automatically discover the hidden patterns and structures from the target DS-IR domain, a model to incorporate these structures to the IR process, and a scoring mechanism for the retrieved documents. We present experiments on two different DS-IR domains: biomedical and patents. We report results for two different evaluation measures, which indicate the effectiveness of the proposed approach. Simple Paraphrase. In the realm of large-scale information retrieval, a crucial challenge lies in developing techniques that can accurately address complex, domain-specific queries. Despite the abundance of datasets and sources available for domain-specific IR, most existing methods rely on supervised machine learning approaches. However, exploring the potential of unsupervised learning techniques in this domain can be a valuable direction. To address this gap, we introduce a novel DS-IR approach that harnesses the power of unsupervised learning to uncover domain-specific patterns and relationships. Our framework comprises three key components: a pattern discovery module that automatically identifies hidden structures in the target domain, a structural integration module that incorporates these patterns into the IR process, and a scoring mechanism that evaluates the retrieved documents. We demonstrate the effectiveness of our approach through experiments on two distinct DS-IR domains: biomedical and patents, using two evaluation metrics. Adversarial Paraphrase. Harnessing the potential of unsupervised machine learning is a critical challenge in the large-scale information retrieval domain, where fine-tuned, domain-specific solutions are desperately needed to effectively answer complex natural language queries. Despite the wealth of readily available datasets for domain-specific IR, many prevailing approaches still rely heavily on supervised machine learning methodologies. Recognizing the untapped potential of unlabeled data, our novel domain-specific IR technique leverages the power of unsupervised learning to distill unique patterns and structures peculiar to specific domains. Comprising three key components - a unsupervised algorithm to automatically unearth these domain-specific patterns, a framework to integrate them seamlessly into the retrieval process, and a scoring mechanism to gauge retrieved document relevance - our approach promises to revolutionize the DS-IR landscape. Experimental evaluations across two distinct domains, biomedical and patents, validate the effectiveness of our proposed technique through robust performance on various evaluation metrics.	– 5 5

Table 14: Examples of original AI texts with their simple and adversarial paraphrases (guided by OpenAI-RoBERTa-Large [30]). GPT-4o quality ratings are provided for each paraphrased version.